4. It is used to calculate the orbital angular momentum using the expression

$$Angular\ momentum = \sqrt{l(l+1)}\,\frac{h}{2\pi} ----(2.14)$$

Magnetic quantum number (m_l):

1. It is denoted by the letter 'm_l'. It takes integral values ranging from -l to +l through 0. i.e. if l=1; m = -1, 0 and +1
2. Different values of m for a given l value, represent different orientation of orbitals in space.
3. The Zeeman Effect (the splitting of spectral lines in a magnetic field) provides the experimental justification for this quantum number.
4. The magnitude of the angular momentum is determined by the quantum number l while its direction is given by magnetic quantum number.

Spin quantum number (m_s):

1. The spin quantum number represents the spin of the electron and is denoted by the letter 'm_s'
2. The electron in an atom revolves not only around the nucleus but also spins. It is usual to write this as electron spins about its own axis either in a clockwise direction or in anti-clockwise direction. The visualisation is not true. However spin is to be understood as representing a property that revealed itself in magnetic fields.
3. Corresponding to the clockwise and anti-clockwise spinning of the electron, maximum two values are possible for this quantum number.
4. The values of 'm_s' is equal to -½ and + ½

Table 2.1 Quantum numbers and its significance

Shell	Principal quantum number (n)	maximum number of electron in a shell (orbital) $2n^2$	Azimuthal quantum number (l) = 0,1.... (n-1)	Maximum no. of electron in a orbital 2 (2l+1)	Magnetic quantum number (m) different possible orientation of orbital	Designation of orbitals in a given shell
K	1	$2(1)^2 = 2$	0	2[2(0)+1] = 2	0	1s
L	2	$2(2)^2 = 8$	0	2	0	2s
			1	2[2(1)+1] =6	-1, 0, +1	$2p_y$, $2p_z$, $2p_x$

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